

TYPED CONTROL, INFORMATION, AND SADDLE ENVELOPES FOR KINETIC COTANGENT PHASES

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ABSTRACT. Exact observations are represented by an arbitrary-codimension stratified current tower. The state retains the unnormalized coarea current, its projective direction, and log evidence, so rare observations do not require a false uniform normalizer bound. A Feller evidence kernel, a uniform coarea coefficient theorem, and a quantitatively reachable chaotic belief cluster yield the exact belief DPP, reduced kinetic game, and statistical tangent.

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1. ROUND-TEN RECONSTRUCTION: STRATIFIED EVIDENCE STATES AND AN EXACT FILTERING GAME

An exact observation need not have a uniformly positive density. The state therefore carries the unnormalized observation current and its evidence; only after this pair is formed is a normalized posterior defined. Exact control is kept on the full belief state. Density reduction is proved only for the quantitatively chaotic belief cluster reachable from the declared preparation and control class.

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1.1. Admissible stratified observation channels. On every chronological hard-sphere cell, an observation channel O_k^u is a smooth map of fixed rank on each member of a finite stratification. Critical strata and their images are retained as lower-rank current components rather than identified with grazing collisions. Repeated observations and collision faces generate the tower

$$\mathfrak{B} = \bigoplus_{r \geq 0} \mathcal{M}_W(\mathcal{S}_r),$$

where $\mathcal{S}_r$ is the union of codimension- r observation/contact strata with its normal orientation and compatibility traces.

Theorem 1.1 (Closed observation–collision current tower). *Free transport, specular reflection, actual-contact creation, prediction under an admissible control, and coarea observation update extend to a positive closed transition operator on $\mathfrak{B}$. If Λ^- is the predicted unnormalized belief and y is observed, the update is the coarea current*

$$\mathcal{O}_{u,y}\Lambda^- = \sum_S 1_{S \cap O_u^{-1}(y)} J_{O_u|S}^{-1} \Lambda^-|_S.$$

Its evidence is $Z(u, y; \Lambda^-) = \langle 1, \mathcal{O}_{u,y}\Lambda^- \rangle$.

Proof. Start from smooth densities on transverse strata. The coarea theorem gives the oriented update and its next trace whenever a level set meets a collision face. The B2 Green graph controls transport and reflection of each normal current; adjoining one observation normal raises codimension by one and the factorial codimension weights make the tower sum bounded. Constant-rank charts cover regular strata, while critical strata are promoted to their own lower-rank components. Closing the graph gives the positive transition operator. No assertion that all observation currents are ordinary B2 interior/binary-flux pairs is made. $\square$

Lemma 1.2 (Evidence lift). *The pair*

$$(\mathcal{O}_{u,y}\Lambda^-, Z(u, y; \Lambda^-))$$

is continuous in the weighted-strict topology on each stratum chart, including sequences with $Z \downarrow 0$. The normalized posterior is used only on $\{Z > 0\}$; risk-sensitive expectations are integrated against the evidence measure $Z(u, y; \Lambda^-)dy$.

Proof. Unnormalized coarea currents vary continuously under constant-rank perturbations and weighted tightness. Their total mass is the same continuous pairing with one. Keeping the pair avoids division by a small number. Bayes' formula integrates a posterior functional as $\int Z(y)F(\Lambda_y/Z(y))dy$, so zero-evidence observations contribute zero and no uniform lower bound is required. $\square$

1.2. The exact belief transition and DPP. A strategy is predictable with respect to the observation history. Prediction uses the B2 source-controlled augmented-law transition; update uses Theorem 1.1.

Theorem 1.3 (Feller evidence kernel and exact belief DPP). *Prediction followed by evidence-bearing observation defines a Feller kernel on*

$$\hat{\mathfrak{B}} = \{(\Lambda, z) : \Lambda \in \mathfrak{B}_+, z \in \mathbb{R}, \langle 1, \Lambda \rangle = e^z\}.$$

For compact relaxed controls, the finite-horizon upper and lower values obey the exact belief-state dynamic programming principle. The state is the full stratified belief/evidence pair, not its one-particle mean.

Proof. The B2 prediction kernel is Feller on energy/evidence sublevels. Coarea continuity from Lemma 1.2 and dominated integration in the evidence measure give Feller continuity of the update. Ionescu–Tulcea constructs the law for every predictable relaxed strategy. Conditioning at a block endpoint, then applying prediction and the new observation update, gives the tower. Compactness and measurable selection give the minimax DPP. $\square$

1.3. A joint coarea coefficient. The declared observation family consists of finite-dimensional affine statistics of the particle path and actual-contact measure on each chronological cell. Their derivatives, together with the B1 constraint mark, have full affine span on a finite family of positive-activity blocks.

Theorem 1.4 (Uniform coarea local coefficient). *On a compact regular source/control chart, the joint variable consisting of B1 shell constraints, exact observations, and bounded path/contact sources has a mixed lattice/continuous local coefficient at its exact finite saddle. For observation values with finite rate,*

$$\frac{1}{\mu_\varepsilon} \log Z_\varepsilon(u, y) \longrightarrow -\mathcal{J}_u(y)$$

uniformly on compact rate sublevels, and numerator/denominator ratios converge to the corresponding tilted stratified posterior kernel. No lower bound on Z_ε uniform over all y is asserted.

Proof. Adjoin the observation coordinates to the B1 good-block sum map. The stated affine-span condition gives a full-rank regenerative block family. The B1 central, lattice minor-arc, and large-frequency estimates therefore extend to the joint Fourier/coarea inversion, uniformly in relaxed controls. Tilting at the exact joint saddle gives the local coefficient and its exponential rate. Applying the same coefficient to a bounded posterior numerator cancels the common local prefactor and gives the ratio theorem. Critical observation strata are treated in their lower-rank chart, not hidden in a billiard singularity shield. $\square$

1.4. Reachable quantitative chaos. Let $\mathfrak{R}_\varepsilon$ be the smallest belief cluster containing the B1 prepared laws and closed under the declared prediction and observation kernels. Let $\kappa_j(\Lambda)$ be its connected correlation currents.

Lemma 1.5 (Propagation of the reachable cluster). *For every fixed horizon and strategy,*

$$\sup_{\Lambda \in \mathfrak{R}_\varepsilon} \|\kappa_j(\Lambda)\|_{\text{trace}} \leq j! C^j \mu_\varepsilon^{1-j}$$

after the finite evidence exponent is included. The estimate is stable under exact observations from Theorem 1.4.

Proof. B1 preparation has the connected cluster estimate. B2 prediction propagates it by the trace renewal norm. Exact observation is an exponential tilt by a finite-dimensional affine statistic followed by a local coefficient ratio; normal convergence of the joint source pressure preserves the connected cumulant scaling. Induction over finitely many observation blocks gives the uniform estimate. Arbitrary beliefs sharing the same one-particle marginal are not included. $\square$

Theorem 1.6 (Reduced kinetic game and statistical tangent). *The exact controlled value is the belief-state value of Theorem 1.3. On the reachable cluster, it converges to the kinetic information-state game obtained from the one-particle density plus the finite evidence/phase coordinates; the error is $o(1)$ uniformly over strategies. At a regular phase, exact finite-centered likelihood ratios obey LAN and the posterior satisfies the corresponding local Bernstein–von Mises theorem.*

Proof. Lemma 1.5 makes all higher connected correlations subleading in the B2 generator and in bounded contact rewards, uniformly over reachable strategies. Apply the B4 nonlinear semigroup limit block by block inside the exact DPP; the evidence and phase coordinates are retained rather than discarded. For LAN, expand the exact finite joint pressure at its exact mean. B3 supplies the joint Gaussian process and positive quotient covariance; the coarea coefficient supplies the local denominator. Le Cam’s third lemma and Laplace expansion give the Gaussian posterior limit. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared model-specific regular regime. Cross-paper inputs are used only in the acyclic order recorded in the round-ten dependency ledger. Repository source, proof-structure, counterexample, and build gates are internal reproducibility checks; they do not replace independent external mathematical review.

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